07

Resource Monitoring

Observing systems with metrics, logs, traces and profiles — and turning signals into action

PCA study series · 10 concept slides · 8 scenario questions

One observability suite, five lenses on a running system

1 Cloud Monitoring

Metrics, dashboards, alerting policies and uptime checks across Google Cloud and beyond.

2 Cloud Logging

Collects, stores, searches and routes logs, including audit logs.

3 Error Reporting

Groups application exceptions and alerts you when new ones appear.

4 Cloud Trace

Follows a request across services to show where latency is spent.

5 Cloud Profiler

Continuously samples CPU and memory to show which code is expensive.

6 How they connect

Each tool links to the others, so you can jump from an alert to the logs and traces behind it.

EXAM LENS Map the symptom to the tool: 'is it up' → Monitoring; 'what happened' → Logging; 'where is it slow' → Trace.

Reliability is defined with SLIs, SLOs and SLAs

Term	What it is	Example
SLI	A measured indicator of user-visible behaviour	Share of requests served in under 300 ms
SLO	Your internal target for that indicator over a window	99.9% of requests under 300 ms over 30 days
SLA	A contractual promise with penalties, set looser than the SLO	99.5% availability or service credits
Error budget	100% minus the SLO — the unreliability you can spend on releases	99.9% over 30 days ≈ 43 minutes

EXAM LENS Monitoring underpins all of SRE. When the error budget runs out, slow releases and fix reliability.

A metrics scope gives one pane of glass across projects

1 Scoping project

The project that hosts dashboards, alerts and uptime checks. Make a dedicated one rather than reuse an app project.

2 Monitored projects

Add other projects to the scope — up to hundreds — and their metrics appear together.

3 Access

IAM on the scoping project controls who sees the combined view. Monitoring roles don't grant access to the resources themselves.

4 Beyond Google Cloud

Metrics from other clouds and on-premises systems can be ingested through agents and integrations.

EXAM LENS 'Central team needs a single view of many projects' → dedicated scoping project with those projects added.

Anatomy of an alerting policy

1

Condition

A metric crosses a threshold, a metric goes absent, or a log-based or PromQL query matches.

2

Duration

How long the condition must hold before firing — filters out brief spikes.

3

Channels

Email, SMS, mobile app, Slack, PagerDuty, webhooks or Pub/Sub.

4

Documentation

Runbook text sent with the alert so responders know what to do.

5

Incident

Opened when the policy fires; closes automatically once the condition clears.

EXAM LENS Alert on symptoms users feel (latency, errors) rather than every internal cause.

What you see with and without the Ops Agent

Built-in VM metrics

Collected from the hypervisor, no install

- CPU utilisation
- Disk read/write bytes and operations
- Network traffic in and out
- Uptime and basic instance status

With the Ops Agent

Installed on the guest OS

- Memory utilisation and swap
- Disk space used per file system
- Process-level metrics
- System and application logs (via Fluent Bit and OpenTelemetry)

EXAM LENS 'Alert on memory usage' or 'disk almost full' for a VM → install the Ops Agent first.

Uptime checks and custom metrics close the gaps

1 Uptime checks

Probe HTTP, HTTPS or TCP endpoints from locations around the world, optionally validating response content. Combine with an alert to page on outages.

2 Private endpoints

Uptime checks can also reach internal resources registered in Service Directory.

3 Custom metrics

Publish business or application measures — queue depth, orders per minute — using OpenTelemetry or the Monitoring API.

4 Autoscale on them

A managed instance group can scale to keep a Monitoring metric, such as backlog per VM, at a target.

EXAM LENS 'Detect an outage the way customers would' → uptime check from multiple regions plus an alerting policy.

Audit and platform logs differ in default and retention

Log type	Enabled	Default retention	Where it lives
Admin Activity audit	Always, free	400 days	_Required bucket
System Event audit	Always, free	400 days	_Required bucket
Data Access audit	Off by default (on for BigQuery)	30 days	_Default bucket
Platform and app logs	On	30 days (configurable)	_Default bucket

EXAM LENS The _Required bucket cannot be changed or deleted. Anything longer or cheaper goes through a sink.

The Log Router decides where every log goes

Sinks match a filter and copy logs to a destination — export for retention, analysis or third-party tools.

Cloud Storage	Cheapest long-term retention; add Archive class and a retention lock for compliance.
BigQuery	SQL analysis of logs; visualise in Looker Studio.
Pub/Sub	Streaming to external systems such as a SIEM (Splunk, Elastic).
Log bucket	Centralise logs from many projects; query with Log Analytics.

Debugging tools for code in production

1 Error Reporting

Aggregates stack traces from supported languages, counts occurrences and notifies on new errors. Use it to know what is breaking right now.

2 Cloud Trace

Distributed tracing with latency reports per endpoint. Use it to find which hop in a microservice chain is slow.

3 Cloud Profiler

Low-overhead continuous profiling of CPU and heap in production. Use it to cut compute cost or fix hot code paths.

EXAM LENS 'Lower overhead in production, find hot functions' → Profiler. 'Which service adds latency' → Trace.

Decision cheat sheet: symptom or need to tool

If the scenario says…	Reach for
Single view across many projects	Metrics scope in a dedicated scoping project
VM memory or disk-space alerts	Ops Agent + alerting policy
Customers report site down; detect it first	Uptime check + alerting policy
Keep audit logs for years, cheaply	Log sink ⟶ Cloud Storage (Archive, locked retention)
Run SQL over logs	Sink to BigQuery, or Log Analytics
Stream logs to an external SIEM	Sink ⟶ Pub/Sub
Find the slow microservice	Cloud Trace
Reduce CPU cost of code	Cloud Profiler



8 PCA-style scenarios

Read every requirement word, eliminate options that break one, then pick the most managed answer that satisfies them all. Each question is followed by its answer and the reasoning.

Question 1 of 8

Log retention

Regulators require all Admin Activity audit logs for the organisation to be kept for 7 years, at the lowest possible cost, and protected from deletion.

A. Rely on the _Required log bucket

B. Create an organisation-level aggregated sink to a Cloud Storage bucket using Archive class with a locked 7-year retention policy

C. Create a sink to BigQuery and set table expiry to 7 years

D. Export logs manually every quarter to an on-premises server

Answer 1: B

B Create an organisation-level aggregated sink to a Cloud Storage bucket using Archive class with a locked 7-year retention policy

Why it fits

An aggregated sink at the organisation captures every project. Cloud Storage Archive is the cheapest durable store, and Bucket Lock prevents deletion.

Why the others don't

A — The _Required bucket keeps logs for 400 days only.

C — Works but costs more for data that is rarely read.

D — Manual, error-prone and outside managed controls.

EXAM LENS Long-term + cheap + immutable logs ⟶ sink to Cloud Storage with retention lock.

Question 2 of 8

VM metrics

Your operations team wants an alert when memory usage on a fleet of Compute Engine VMs exceeds 85%. The metric does not appear in Cloud Monitoring.

A. Enable the Compute Engine API again

B. Install the Ops Agent on the VMs and create the alerting policy

C. Create an uptime check for each VM

D. Turn on Data Access audit logs

Answer 2: B

B Install the Ops Agent on the VMs and create the alerting policy

Why it fits

Memory is measured inside the guest OS, which the hypervisor cannot see. The Ops Agent collects it and makes the metric available for alerting.

Why the others don't

A — The API is clearly enabled already; that does not add guest metrics.

C — Uptime checks test reachability, not memory.

D — Audit logs record API access, not resource usage.

EXAM LENS Memory, disk-space or process metrics on VMs → Ops Agent.

Question 3 of 8

Metrics scope

A central SRE team must build dashboards and alerts covering 40 application projects without being granted access to change resources in them.

A. Grant the SRE team Owner on all 40 projects

B. Create a dedicated scoping project, add the 40 projects to its metrics scope and grant the team Monitoring roles there

C. Export every project's metrics to BigQuery nightly

D. Copy the dashboards into each project

Answer 3: B

B Create a dedicated scoping project, add the 40 projects to its metrics scope and grant the team Monitoring roles there

Why it fits

A metrics scope aggregates metrics from many projects into one place, and Monitoring roles on the scoping project give visibility without resource permissions.

Why the others don't

A — Violates least privilege.

C — Not real-time and no alerting.

D — Forty separate views, not one pane of glass.

EXAM LENS 'Many projects, one view' → metrics scope.

Question 4 of 8

Latency

Users report slow checkout. The request passes through five microservices on GKE. You need to identify which service adds the most latency.

A Cloud Profiler

B Error Reporting

C Cloud Trace

D Uptime checks

Answer 4: C

C Cloud Trace

Why it fits

Distributed tracing records the time spent in each service along one request path, pinpointing the slow hop.

Why the others don't

A — Shows expensive functions inside a service, not cross-service latency.

B — Groups exceptions; slow is not an error.

D — Tells you if the endpoint is up, not where time goes.

EXAM LENS 'Which service is slow' → Trace. 'Which function is expensive' → Profiler.

Question 5 of 8

Log export

The security team runs Splunk on-premises and needs Google Cloud audit logs delivered continuously in near real time.

A Create a log sink to Pub/Sub and have Splunk ingest from the subscription

B Create a log sink to Cloud Storage and download hourly

C Give the Splunk server a service account key with Logging Viewer

D Email the logs daily using a scheduled query

Answer 5: A

A Create a log sink to Pub/Sub and have Splunk ingest from the subscription

Why it fits

A Pub/Sub destination streams matching logs as they arrive; Splunk (directly or through a Dataflow template) consumes them continuously.

Why the others don't

B — Storage sinks write in batches; not near real time.

C — Polling with long-lived keys is insecure and inefficient.

D — Daily and manual.

EXAM LENS Streaming to third-party tools → Pub/Sub sink.

Question 6 of 8

Outage detection

Your public API is served from three regions. You want on-call engineers paged through PagerDuty within minutes if customers anywhere can't reach it.

A. Uptime check against the public endpoint from multiple locations, with an alerting policy using a PagerDuty channel

B. A log-based metric on HTTP 500 responses

C. Cloud Profiler on every backend

D. A billing budget alert

Answer 6: A

A Uptime check against the public endpoint from multiple locations, with an alerting policy using a PagerDuty channel

Why it fits

Uptime checks test the service from outside, as customers do, and the alerting policy routes failures to PagerDuty.

Why the others don't

B — If requests never arrive, no 500s are logged.

C — Profiles performance, does not detect outages.

D — Unrelated to availability.

EXAM LENS 'From the user's perspective' → uptime checks.

Question 7 of 8

Cost of code

A Java service's compute bill has doubled since the last release. You need to find which methods use the most CPU in production with negligible overhead.

A Cloud Trace

B Cloud Profiler

C Ops Agent process metrics

D Data Access audit logs

Answer 7: B

B Cloud Profiler

Why it fits

Profiler continuously samples CPU and heap in production and attributes usage to specific methods with very low overhead.

Why the others don't

A — Shows request latency, not per-method CPU.

C — Shows a process is busy, not which code.

D — Records who accessed data.

EXAM LENS 'Which code consumes CPU or memory' → Profiler.

Question 8 of 8

SRE

Your SLO is 99.9% availability measured over 30 days. By day 20 the service has been down for 40 minutes in total. What does SRE practice suggest?

A. Nothing — the SLA is 99.5%, so there is plenty of room

B. The error budget (about 43 minutes) is nearly spent: prioritise reliability work and slow risky releases

C. Raise the SLO to 99.99% to motivate the team

D. Remove the SLO since it is being missed

Answer 8: B

B The error budget (about 43 minutes) is nearly spent: prioritise reliability work and slow risky releases

Why it fits

99.9% over 30 days allows roughly 43 minutes of downtime. With 40 used, the remaining budget is tiny, so risk should be reduced.

Why the others don't

A — The internal SLO, not the SLA, guides engineering decisions.

C — A stricter target is already more likely to fail.

D — Discarding the SLO removes the signal.

EXAM LENS Error budget = (1 − SLO) × window. Spend it on change; protect it when low.